

Dylan Yost-Slinker

Wayne State University Chemistry 5101 Cass Ave. Detroit, Michigan

Personal Summary

PhD candidate with 3 years of experience in computational chemistry in determining the impact of pH on fluorescent protein structure and function. Specializing in constant pH molecular dynamics simulations, coding, protein modeling, drug design, and tool development. Focused on developing automated and efficient workflows to aid in protein engineering.

Education

2023 – Present : Graduate Student Researcher

Wayne State University, Detroit, MI

GPA: 3.95/4.00

Honors: Norman A. LeBel Endowed Graduate Award in Organic Chemistry, CBI Scholar Award, Mikoryak Award

Relevant Coursework: Advanced Bio-organic Chemistry and Drug Design, Metabolism, Computational Chemistry.

Focus: Determining the mechanism of pH sensitivity in pH-sensitive fluorescent proteins. **Key Research**

Interests: Molecular Dynamics, Automation, Tool Development.

2019 – 2023: BS in Chemistry

University of California, Merced, Merced, CA

GPA: 3.92/4.00

Honors: Dean's List

Relevant Coursework: Organic Spectroscopy and Computation, Chemical Thermodynamics and Kinetics, Probability and Statistics, Engineering Computing

Projects

May 2023 – : Graduate Student Researcher

Supervised by: Professor Alice Walker

- Developed a suite of tools to aid in analyzing constant pH simulation data, including an automated workflow.
- Investigated the mechanics behind pH sensitivity in pH-sensitive fluorescent proteins and designing new fluorescent proteins.

Academic Publications

Synthetic Heparan Sulfate-Like Oligosaccharides as Tools to Decipher Growth Factor Recognition.

Under review.

Wakpal, Joseph; Singh, Kartikey; Hotor, Michael; Philip, Livia; **Yost-Slinker, Dylan**

Skills

Python, Linux (Ubuntu), AutoParams, Amber, Automation, Molecular Dynamics, Constant pH simulations
Languages: English (Native)

References

Available upon request